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1. Getting Started

1.1 What This Software Does

Praveen Garments Billing / POS / Inventory is the private, single-store business system that runs the daily shop counter. It covers the complete garment trade workflow: catalogue with size and colour variants, barcode and QR identification down to the exact variant, stock tracked per location, fast POS billing with cash, UPI, card and credit payments, printing of GST invoices, customer and supplier accounts, purchases and returns, exchanges, day-to-day expenses, and seventeen management reports with a live dashboard. Every rupee and every unit of stock is written to a PostgreSQL database protected by row-level security, so the numbers on the dashboard are the same numbers an auditor would find in the ledger.

The system is deliberately private: there is no public sign-up, no self-service registration, and no data shared with third parties. User accounts are created by an administrator inside the shop, and every sensitive action is permission-checked in the database itself, not only hidden in the interface. This guide is the single official reference for the store team and for anyone testing the system before acceptance. It is a living document: whenever a feature changes, the relevant section must be updated in the same release so the manual and the software never drift apart.

1.2 System Requirements

The application is a web system and needs no installation on the billing computer. Open it in a modern Chromium-based browser (Chrome or Edge, current version). Each counter station should have a stable internet connection because the database is hosted online; billing stops safely with an error message if the connection drops, and it never records a payment it could not finish. Recommended screen width is 1280 px or more for the POS screen, although the layout adapts down to mobile widths. Optional hardware — a USB barcode scanner, a label printer and a receipt printer — connects to the same computer and needs no special drivers; see the Hardware Setup chapter for honest details of what has and has not been physically tested.

1.3 Logging In

Open the shop URL and you will see the sign-in page with the shop name and logo. Enter the email address and password given by the administrator. If the password is wrong the page shows a clear inline message; if the account has been deactivated, the system signs you out with an explanation. After signing in you land on the Dashboard. Use the account menu at the top right to sign out; the action is recorded in the audit trail. Password resets are done by an administrator from the Users page, or through the forgot-password link which emails a reset link to your registered address.

1.4 Users, Roles and Permissions

Every person who uses the system gets one account with one role. The role decides which pages appear and, more importantly, which operations the database accepts. Four roles exist: **admin** (full control including

users and settings), **manager** (everything commercial — purchases, returns, approvals, reports), **staff** (catalogue and stock work, no approvals), and **cashier** (POS billing and held bills only). Permission names follow the action, for example `manage_products`, `manage_inventory`, `view_reports`, `manage_expenses`.

Capability	Admin	Manager	Staff	Cashier
Create / edit users and roles	Yes	No	No	No
Company settings & branches	Yes	No	No	No
Create products, variants, barcodes	Yes	Yes	Yes	No
Stock adjustments & transfers	Yes	Yes	Yes	No
POS billing & held bills	Yes	Yes	Yes	Yes
Approve expenses, refunds, returns	Yes	Yes	No	No
Purchases, suppliers, payables	Yes	Yes	No	No
View reports & audit trail	Yes	Yes	No	No

Table 1-1 — Role capability matrix (enforced in the database, not just hidden in the UI).

Hiding a button is not security: the enforcement happens in the database. A cashier who somehow calls an admin API directly receives an authorization error, because the permission check runs again server-side inside the transaction. The role matrix itself is read-only in Settings; changing a role's permissions is a database-level change reserved for future phases, and the screen says so honestly.

1.5 A Quick Tour of the Screens

Menu	What you do there
Dashboard	Today's sales, bills, items sold, payment split, low stock, dues and expenses at a glance.
POS	The billing counter: scan, search, cart, customer, discount, checkout and print.
Sales	Bill history, invoice reprint, returns, exchanges and cancellations.
Payments	Every money movement: sales payments, customer receipts, supplier payments, refunds.
Products	Catalogue: products, variants, SKU, barcode, QR, images, attributes.
Inventory	Current stock by variant and location, adjustments, transfers, history, locations.
Purchases	Purchase orders, supplier invoices (goods receipt), supplier returns.
Customers / Suppliers	Accounts, statements, outstanding balances, payment history.
Expenses	Shop spending with categories, receipts and approval flow.
Reports	Seventeen reports with date filters and CSV export.

Menu	What you do there
Settings / Users	Company profile, branches, prefixes, business rules, user accounts.

Table 1-2 — The main menu in daily-use order.

2. Dashboard and Reports

2.1 Reading the Dashboard

The Dashboard opens with a period selector (today, yesterday, this week, this month, or a custom range). Every card is computed from real transactions in that period: gross sales, net sales, number of bills, items sold, average bill value, payment-method split, refunds, expenses, profit estimate, low-stock count and out-of-stock count. Cards that would be empty show an honest zero — the system never displays fake or sample data. If the reporting functions are ever not yet applied on the database, the page says so plainly instead of showing wrong numbers.

Reconciliation habit for the owner: at closing time, compare the payment split card with the cash drawer count and the UPI statement. The numbers must match to the rupee because both come from the same sale payments table. If they differ, use the Payments page to find the transaction that explains the gap before assuming an error — for example a bill paid half in cash and half by UPI appears in both columns.

2.2 Reports Overview

The Reports page lists seventeen reports, each gated by the `view_reports` permission. Every report supports a date range, applies filters only when you press Apply, and can be exported to CSV (capped at 5,000 rows to protect the browser). The most used daily reports are Sales, Payments, Cash summary and Returns; GST and Profit reports are designed for the accountant at month end. Values shown are recomputed directly from bills, items, payments and purchase lines — the audit found every report in agreement with the underlying tables.

Report	Answers the question
Sales	How many bills, how much gross/net, tax and discount, per day — filter by payment status.
Product sales	Which variants sold how many units at what value in the period.
Catalog performance	Which products move fastest and which sit on the shelf.
Payments	Money in by method, day by day, including dues collected.
Cash	Opening float, cash sales, refunds paid, expenses — the drawer story.
GST / Tax	Taxable value and CGST / SGST / IGST split per rate.

Report	Answers the question
Profit	Sales margin minus weighted-average purchase cost.
Stock valuation	What the current stock is worth at cost and at price.
Stock performance	Turnover and dead stock by variant.
Purchase / Supplier	Purchases, payables and supplier activity.
Customer	Top customers, dues and payment behaviour.
Expense / Returns	Spending by category; returns and refunds.
Audit trail	Who changed what, when — every sensitive action.

Table 2-1 — The seventeen reports (grouped) and their business use.

3. Product Catalog

3.1 Creating a Product

Open Products and press New Product. The wizard has five short sections: basics (name, product code, description, gender, fabric), classification (category, brand), pricing (MRP and selling price, GST rate, HSN code), and variants. The product code becomes part of every SKU, so keep it short and stable, for example SHIRT for shirts. Prices entered at product level act as defaults that each variant can override — a variant with no price of its own sells at the product price, and changing the product price later only affects variants that never had their own price set.

Categories can be nested one level (Shirts under Men), brands are flat, and sizes and colours are shared lists; manage them from Products, then Attributes, where you can add, rename and deactivate entries without deleting history. Deactivating an attribute hides it from new variants but leaves old bills untouched.

3.2 Variants: Size × Colour

Garments sell by exact variant, so the wizard builds a matrix: pick the sizes and colours the product comes in and the system creates one variant per combination — Blue/M, Blue/L, Black/M, Black/L each get their own row. Each variant can carry its own selling price, MRP, barcode and QR code. Deactivating a variant hides it from POS search but keeps its sales history; you can reactivate it later. You cannot delete a variant that has ever moved stock, which protects the ledger; deactivation is the correct way to retire it.

3.3 SKU, Barcode and QR — How Identification Works

Every variant has three identifiers and all three resolve to that exact variant, never to the parent product and never to a sibling variant. The SKU is built from the product code plus size and colour (for example PG-SHIRT-BLU-M) and is unique case-insensitively across the catalogue. The barcode is an EAN-13

number generated with a valid checksum using the reserved in-store prefix range 20000000–29999999, or you can type your supplier’s barcode if the garment already carries one. The QR code is a short store-issued text identifier. Duplicate barcodes and duplicate QR codes across different variants are rejected by the database, so one scanned code can never mean two products.

Why exactness matters at the counter

A scanned barcode must add the precise variant to the cart — Blue/M, not Blue/L. The POS scanner endpoint matches the identifier to a single variant row and replies with that variant’s price, stock and location. Partial matches are never auto-added: typing a fuzzy product name shows a search list instead, and only an exact SKU / barcode / QR / product-code match adds directly to the cart.

Print labels from the Labels page: choose a template size, generate the barcode and QR images, and print through the browser print dialog on a label printer or normal printer. Scan one printed label back with the counter scanner before trusting a whole batch — if it opens the exact variant on the product screen, the labels are correct.

3.4 Product Images

Upload a PNG, JPEG or WebP image up to 2 MB on the product page; SVG is rejected for safety. The image lives in a private Supabase storage bucket with strict upload policies, and only users with the `manage_products` permission can add, replace or remove it. Replacing keeps the same display position so the catalogue does not jump around. Invalid files and oversized files are rejected with a clear message both in the browser and again on the server, so a rejected image never reaches storage.

4. Inventory Management

4.1 Stock Model: Variant × Location

Stock is tracked per variant per location. A shop typically has one Main store; you can add a Warehouse or shelf locations from Inventory, then Locations (admin permission). Selling from the POS deducts the stock of the selected location; receiving a purchase adds to it. Every change writes an append-only movement row in a ledger that nobody — not even an admin — can edit or delete from the interface. The current balance is always the sum of the movements for that variant and location, and the audit verified this equality across the whole ledger.

4.2 Opening Stock, Adjustments and Transfers

When a product first arrives, record Opening Stock once per variant and location — the system refuses a second opening for the same pair, because opening twice would double-count. Daily corrections use Adjust: choose the variant, location, new quantity or delta, and a reason. Reasons are mandatory for adjustments, damage and loss, and damage movements never increase sellable stock. Moving goods between locations uses

Transfer, which books a transfer-out and a transfer-in in one atomic step — both sides appear immediately in the history of both locations, and a half-done transfer cannot exist.

Movement type	Written when	Effect
OPENING	First opening stock of a variant at a location	Sets the baseline quantity (once only)
PURCHASE	Supplier invoice received (full or partial)	Stock increases by received quantity
SALE	POS checkout completes	Stock decreases by billed quantity
SALES_RETURN	Customer returns a garment (resellable)	Stock increases back
DAMAGE_RETURN	Customer returns a damaged garment	Not added to sellable stock
PURCHASE_RETURN	Goods sent back to supplier	Stock decreases
ADJUSTMENT	Manual correction with reason	Sets or shifts quantity
TRANSFER_OUT / IN	Location-to-location move	One atomic pair of rows
EXCHANGE_OUT / IN	Exchange of one variant for another	Old stock back in, new stock out

Table 4-1 — The stock ledger movement types and when they are written.

4.3 Low Stock and Out of Stock

Set a reorder level per variant (the Inventory page shows a reorder column and a low-stock badge when quantity falls to or below it). The Dashboard low-stock and out-of-stock cards use the same thresholds. Selling stops safely at zero: when the last unit is sold the balance becomes exactly 0, and the next sale attempt fails with `INSUFFICIENT_STOCK` — the database guard, not just the interface. Negative stock can be switched on in Settings only if the owner explicitly accepts that business rule; by default it is off.

4.4 Reading Stock History

The History tab shows the movement ledger in newest-first order with an older / newer pager that pages through the ledger without ever loading it all at once. Each row shows the movement type, quantity with sign, resulting balance, the document that caused it (bill number, purchase invoice, return or exchange reference), the location and the user. This is the screen an auditor asks for first: every balance on the Current tab can be traced line by line to its movements.

5. POS — Billing at the Counter

5.1 Starting a Bill

The POS screen is the daily workhorse. The scan box at the top takes keyboard input from the barcode scanner or typing. Pressing Enter after a code looks it up: an exact match on SKU, barcode, QR or product code adds the item straight to the cart with its price and current stock; anything else runs a product search listing matches by name. Scanning the same item again raises the cart quantity — scan three times, quantity three — and the audit verified the database ends with exactly three units deducted and one sale row, never three sales.

Pick the billing location if you run more than one (it defaults to Main). Adjust quantities with the number controls, remove a line with its remove button, and clear the whole cart with one action. A bill note can be attached and is printed on the invoice. Pressing Hold parks the cart as a held bill you can resume later from the held-bills list; discarding a held bill writes no financial record at all.

5.2 Attaching a Customer and Discounts

Choose an existing customer by searching name or phone, or press new customer to create one inline with just a name and phone number. A walk-in sale needs no customer. Line-level discounts can be applied per item, and the cart footer shows subtotal, discount total, GST and grand total live. GST is computed from each variant's rate stored on the bill line, so a mixed cart of 5% and 12% items splits the tax correctly. Prices and taxes are snapshotted onto the bill at the moment of sale, so later price changes never rewrite an old invoice.

5.3 Completing the Sale

Press Charge to open the checkout dialog. Pick the payment method (Cash, UPI, Card, Credit as enabled in Settings), enter the amount taken, and for cash see the change due instantly — a bill of 1,250 paid with 2,000 shows change 750. Split payment is supported: part cash and part UPI on one bill, each with its own reference. Press Complete Payment once; the button disables itself while the request runs, and the database transaction creates the sale, its items, its payments and the stock deductions together or not at all. A failure mid-way shows an error and leaves no half-bill behind. On success the invoice opens ready to print.

6. Customers and Credit Accounts

6.1 Customer Records

The Customers page holds every buyer the shop bills by name. Create a customer with name and phone (phone is the practical identifier at the counter), and optionally email, GSTIN for business customers, billing address and state (used for IGST), and a retail/wholesale type. Search covers name, phone, email and GSTIN. Editing a customer never rewrites past bills — the invoice keeps the customer name snapshot taken at sale time, exactly like prices. Deactivating a customer hides them from POS selection while keeping their full history.

6.2 Credit Sales and Dues

A credit sale is simply a bill paid partly or fully by the Credit method: pay 2,000 on a 5,000 bill and the customer's due becomes 3,000. Dues live on the bill, not on a guessed balance field — the customer's outstanding is always the sum of unpaid bill portions, minus credits from returns and payments, and the audit reconciled this sum to the rupee for every test account. The Customers page shows each account's current dues; the dashboard pending-payments card aggregates the same figure for the whole book.

6.3 Receiving Payments and Statements

Open the customer's page and press Record Payment to take money against dues. The payment is allocated to outstanding bills oldest-first (FIFO), prints a receipt with its own CR number, and appears on the Payments page. Paying 1,000 of the 3,000 due leaves 2,000 — check it on the statement. Press Statement to see the full ledger for any date range: bills, till payments, receipts, return credits and cancellations, each with a running balance, ending on the same due figure the customer page shows. If the two ever disagree, trust the statement and report it — the audit found no such disagreement on any account.

7. Checkout and Payment Methods

7.1 Method Rules

Method	Works like	Watch out for
Cash	Money in the drawer; change is computed for you	Count the change shown on screen, it is exact
UPI	Digital transfer; optional reference note	Confirm the UPI app credited before completing
Card	Swipe / tap on the EDC machine; optional reference	The bank slip number is handy as the reference

Method	Works like	Watch out for
Credit	Dues recorded on the customer's account	Only works when a customer is attached to the bill
Split	Any two methods on one bill	Total of parts must equal the bill total

Table 7-1 — Payment methods and their rules.

The checkout refuses impossible math: the parts of a split payment must sum to the bill total, credit is rejected without a customer, and change is only shown for cash. Every payment row stores its method, amount and optional reference, so the Payments report can reconstruct the day exactly as the drawer and bank statements saw it.

7.2 Idempotency — Why Double-Clicks Are Safe

The Complete Payment button disables itself the moment it is pressed and stays disabled until the server replies. Behind the scenes the checkout is one atomic database transaction: sale, items, payments and stock movements commit together or roll back together. Rapid double-clicks therefore cannot create two bills — the audit hammered this path with parallel requests and the ledger stayed consistent: one bill, one payment set, one deduction set, always. If the network drops mid-payment the screen shows an error, nothing is recorded, and the cart is still there to retry.

8. Invoices and Printing

8.1 What Is On the Invoice

Each completed bill gets the next invoice number from your configured prefix (for example INV-2026-000123 — the prefix and year are set in Settings). The invoice prints the company name, logo, address and GSTIN from Settings, the invoice number, date and time, cashier name, customer name and phone when attached, and for every line: product, size and colour, SKU, quantity, MRP, unit price, discount, GST rate and amount, and line total. The footer shows subtotal, discount, tax and grand total, the payments received with methods and references, change given for cash, dues for credit, and the bill note if any. All values are snapshots — a price change tomorrow does not move yesterday's invoice.

A4 and thermal printing

Printing uses the browser print dialog. For A4 the layout fits a standard sheet with margins; for thermal receipt printers choose the printer's paper size in the browser print dialog. Direct ESC/POS printer integration is NOT implemented — printing goes through the browser, which works reliably but is a browser print, not a printer driver integration. Say it honestly: browser printing verified; direct thermal integration not implemented.

8.2 Reprints and History

Open Sales, find the bill (search by invoice number, customer or cashier), and press print to reprint the same invoice document. A reprint is identical to the original — same number, same values — and creates no new financial record. Cancelling a bill (manager permission) asks for a reason, marks the bill CANCELLED, puts sold stock back, and records the action in the audit trail; cancelled bills are excluded from sales totals but remain visible in history so the number sequence stays explainable.

9. Purchases, Suppliers and Stock-In

9.1 Suppliers and Purchase Orders

Suppliers are created with name and phone, optionally GSTIN, address and payment terms. A Purchase Order records what you intend to buy — supplier, location, expected date and lines with variant, quantity and unit cost — and gets the next PO number from your prefix. A PO is just intent: it moves no stock and owes nothing until you receive against it. Purchase history, payables and payments per supplier live on the supplier's page.

9.2 Receiving Goods — the Purchase Invoice

Receiving is the money step: press Receive Goods / Purchase Invoice from the Purchases page. Link the purchase order to pre-fill its pending lines, or leave it unlinked for a direct purchase. Enter the supplier's invoice number and date as printed on their bill. For a linked order the received quantity of each line is capped at what remains pending — ordered 100, already received 60, and the next delivery can receive at most 40; receiving 60 adds exactly 60 to stock, never 100. When you press Receive goods, one atomic transaction books the stock in, creates the payable to the supplier, and advances the linked PO — there is no half-received state. Uncheck Receive goods now to save a draft that touches nothing until confirmed.

Partial receiving rule

Ordered = 100, received = 60 means stock +60 and 40 still pending on the order. The audit verified the database enforces this cap line by line, so over-receiving a linked order is impossible through the interface.

9.3 Supplier Payments and Returns

Pay a supplier from their page with Record Payment; payables are the sum of unpaid invoice amounts, oldest invoices settled first. A Purchase Return sends goods back to the supplier: choose the invoice, return up to the received quantity per line, and the system reduces stock, writes the movement, reduces the payable, and refuses to return more than you actually received. Returns cannot exceed eligible quantity — the guard is in the database.

10. Sales Operations — Returns, Exchanges, Refunds

10.1 Sales Return

A customer brings back a garment: open the bill from Sales and press Return. Each line shows the eligible quantity (bought minus already returned). Return one of three sold and the net effect is exactly minus two on that variant's stock — one SALE row of -3 and one SALES_RETURN row of +1 in the ledger. For a resellable garment the stock comes back as sellable; mark the return as damaged instead and the unit does NOT re-enter sellable stock, it is written off the sellable balance as damaged. The refund choice mirrors how the customer is compensated: cash out of the drawer (recorded as a refund payment), UPI back, or credit onto the customer's account which reduces their dues.

Verified arithmetic

Sell 3, return 1 (resellable): ledger shows SALE -3 and SALES_RETURN +1, net stock -2, refund recorded once. Damaged returns never inflate sellable stock. Both behaviours are enforced by the database engine and were re-verified in the Phase 7 regression.

10.2 Exchange

An exchange swaps one variant for another on the same bill. Old garment 999, new garment 1,199: the customer pays the 200 difference through any payment method; new cheaper (1,199 old, 999 new) and the system refunds the 200 the same way. Stock follows the swap — old variant back in, new variant out — and the exchange document records both sides, the price difference and the payment, with its own EX number. The Payments and Cash reports include exchange collections and exchange cash-out so the drawer still reconciles at closing.

Refunds always attach to a return or an exchange document — there is no free-floating refund, so money out of the drawer is always explainable by a document a manager can open.

11. Expenses

11.1 Recording Shop Spending

Cashiers never see this module; it needs the manage_expenses permission. Record an expense with category (manage the category list from the page — Rent, Electricity, Tea, and so on, configurable, never hardcoded), description, amount, payment method, date and optional branch. When you save, the expense sits in a pending state until an approver reviews it — this mirrors how real shops control petty spending. Approving finalises it into the reports; cancelling (with reason) removes it from totals but keeps it visible in history with

the cancellation reason, so the audit trail explains every number the Cash and Expense reports show.

11.2 Receipts and Attachments

Attach the physical receipt as a photo or PDF: PNG, JPEG, WebP or PDF up to 5 MB, validated in the browser and again on the server, stored in a private bucket that only expense-approvers can read. If the attachment upload fails after the expense saved, the system says so honestly — the expense is saved and the receipt can be re-attached from the expense detail. The Cash report subtracts approved expenses from cash, so the drawer story stays complete.

12. Settings and Security

12.1 Settings That Change Behaviour

Settings are grouped into Company (name, logo, address, GSTIN, currency display, timezone), Branches, and business-rule tabs. Every value is stored in the database — nothing about your shop is baked into the code — and a saved setting changes real behaviour, not just the screen: invoice prefixes feed the numbering, low-stock thresholds drive the dashboard badges, and the payment-methods list controls which buttons the checkout offers. After changing anything, save, refresh and check the behaviour in the relevant module — that is exactly what the Phase 7 regression re-tested.

Setting	Where it bites
Company profile & logo	Login page, printed invoices, headers everywhere
Invoice / document prefixes	INV, PO, PI, PR, SR, EX, EXP, CR, SP numbers
Payment methods list	Which methods the checkout and expense forms offer
Low stock threshold	Dashboard low-stock card and Inventory badges
Allow negative stock	Whether the last unit can be oversold (default off)
Return window (days)	How old a bill may be for returns and exchanges
Tax settings	GST rates offered on products and GST report grouping

Table 12-1 — Key settings and where they change behaviour.

12.2 Security Model in Plain Words

All data lives in Supabase PostgreSQL with row-level security (RLS) on every table. In practice this means: signed-in users only ever see rows their role permits, writes go through stored functions that check permission again inside the transaction, the append-only stock ledger cannot be edited by anyone including admins, and the anonymous public key can read nothing of your business data. The audit tried direct table

access, forged requests and wrong-role operations — the database refused them all. Two operational rules keep this true: public sign-up must stay disabled at the Supabase project level (new staff are created by your admin), and the service-role key must never leave the server configuration.

13. Hardware Setup — Honest Status

13.1 What Has Been Verified and How

This section separates three levels of truth: **software-verified** (the workflow works through the interface), **physically verified** (tested with the actual device), and **not tested**. The audit ran all software workflows against the live database; no physical scanner, label printer or mobile device was attached to the audit environment, so this manual claims physical compatibility only where the interface design guarantees it and labels everything else clearly. Never let documentation claim more than was tested.

Device	Status	What this means for you
USB / Bluetooth keyboard-wedge barcode scanner	SOFTWARE VERIFIED (not physically tested)	Scanners that type characters + Enter work by design; plug in and scan into the POS box, then run the 5-minute check below
2D QR scanner (keyboard-wedge mode)	SOFTWARE VERIFIED (not physically tested)	Same as above: it must be in keyboard (HID) mode, not serial mode
Mobile camera QR scanning	IMPLEMENTED, not physically tested	Requires HTTPS and camera permission; message explains fallbacks when unsupported
Receipt / label printing	BROWSER PRINT verified	Print via the browser dialog on any installed printer; no direct ESC/POS integration
Cash drawer (automatic open)	NOT IMPLEMENTED	The drawer is opened manually with the key; nothing in the software triggers it

Table 13-1 — Hardware status: software-verified vs physically-verified vs not tested.

13.2 Installing a Barcode Scanner (5 Minutes)

- Plug the USB scanner into the billing computer (or pair it over Bluetooth) and let it install as a keyboard — no driver needed.
- Open the POS page and click once in the scan box so it has focus.
- Scan any garment label: the code should appear in the box and press Enter automatically (most scanners send Enter after the code; if not, configure the scanner's "Enter suffix / carriage return" option per its manual).
- The exact variant must appear in the cart — if a different size or colour appears, stop and check that label; report it before billing.

- Scan the same item twice: quantity becomes 2. That is the correct behaviour.

Focus note: the scan box keeps focus after each successful add, so continuous scanning works without touching the mouse. If you click elsewhere the scan goes nowhere — click back into the box and the first scan after that lands normally. This is standard keyboard-wedge behaviour, not a fault.

13.3 Mobile Camera QR

On a phone, open the POS page over HTTPS and the camera-scan button becomes available; it asks for camera permission and then scans QR labels live. If the browser reports that camera scanning is not supported, the message explains the two usual causes — the site is not on HTTPS, or the browser lacks the BarcodeDetector capability — and points to the fallbacks, which always work: type or paste the QR text into the scan box, use product search, or use the USB scanner. No physical phone test was performed by the audit, so treat mobile camera scanning as implemented-but-unverified until you try one bill on the shop phone.

13.4 Printers and the Cash Drawer

Install any Windows printer as usual and choose it in the browser print dialog when printing invoices or labels; thermal receipt printers work with their own paper size selected in the same dialog. There is no direct ESC/POS or network-printer integration in this version — that is a future feature, listed honestly in the recommendations chapter. The cash drawer is not connected to the software at all: open it manually when giving change, and reconcile it with the Cash report at closing.

14. Troubleshooting and Common Errors

Symptom	Meaning	What to do
"INSUFFICIENT_STOCK" when billing	The last unit was sold moments ago by another counter	Refresh the cart; the stock guard protected both bills — exactly one sale went through
Login says account disabled	An administrator deactivated the account	Ask the admin to reactivate it from Users
Page shows "Something went wrong"	An unexpected error — the page stopped rather than showing wrong numbers	Reload the page; if it repeats, note what you clicked and tell the administrator (after the Phase 7 fix the known dropdown crash is gone)
Scan adds nothing	Scanner focus lost, or code has no Enter suffix	Click into the scan box; configure the scanner Enter suffix
Camera QR says not supported	Not HTTPS, or browser lacks BarcodeDetector	Use the HTTPS link, update the browser, or use manual entry / search
Image upload rejected	Not PNG/JPEG/WebP or over 2 MB	Convert or shrink the photo and retry

Symptom	Meaning	What to do
Report will not load	Migration not applied or filter too narrow	Check the setup notice on the page; widen the date range
Checkout says network error	Connection dropped mid-payment	Nothing was charged — check Sales to confirm, then retry the bill

Table 14-1 — First-line troubleshooting.

Golden rule: the system never records a payment it cannot finish. If any error appears during money steps, verify on the Sales or Payments page what actually landed, then retry. Partial writes are blocked by design — the audit verified transaction atomicity under deliberate failure injection.

15. Tester / UAT and Production Acceptance

15.1 How To Run a UAT Session

Test with a small, real batch of garments and a second person playing customer. Use statuses honestly: PASS, FAIL, PARTIAL, BLOCKED, NOT TESTED — never "looks fine". For every test write down what you did, what you expected, what actually happened, and the evidence (invoice number, movement row, report figure). The stock arithmetic tests below are the heart of acceptance: the database, not the screen, is the judge.

15.2 Core UAT Checklist

#	Test	Expected result
1	Create product with 4 variants (Blue M/L, Black M/L), barcodes + QR	All four variants listed; each identifier resolves to its exact variant
2	Set opening stock 10 on Blue M; sell 1 over POS	Database stock 9; ledger row SALE -1 referencing the bill
3	Scan same barcode 3 times, complete sale	Cart quantity 3; stock -3 exactly; one bill, one payment set
4	Sell 1 each of Blue M / Blue L / Black M	Each variant -1; no other balance changes
5	Sell with stock 0	Blocked with INSUFFICIENT_STOCK; no negative stock, no invoice
6	Two bills for the same last unit at once	Exactly one succeeds; other fails cleanly; stock ends 0
7	Rapid double-click Complete Payment	One sale, one invoice, one payment, one deduction
8	Cash 2,000 on a 1,250 bill	Change 750 on screen and printed invoice
9	Split 1,000 cash + 1,500 UPI on 2,500	Paid 2,500, due 0; two payment rows with methods

#	Test	Expected result
10	Credit sale 5,000, pay 2,000	Customer due 3,000; pay 1,000 more later, due 2,000; statement closing matches
11	Purchase order 100, receive 60	Stock +60; order shows 40 pending; payable = invoice value
12	Return 3 of a 3-sold item as 1 resellable	Net stock -2; refund recorded once; damaged return adds no sellable stock
13	Exchange 999 item for 1,199 item	Customer pays 200; old stock back, new stock out; EX document present
14	Sell at 999, change price to 1,199	Old invoice still prints 999 with its snapshots
15	Cashier tries Reports / Users / Expenses URLs	Permission denied — database-level, not just hidden buttons
16	Record expense with receipt, approve it	Pending then approved; appears in Expense and Cash reports
17	Change invoice prefix in Settings	Next bill uses the new prefix; old numbers unchanged
18	Full store-day simulation (Part 56 of the audit)	Dashboard, reports, drawer and ledger all reconcile

Table 15-1 — The 18 core UAT checks (statuses: PASS / FAIL / PARTIAL / BLOCKED / NOT TESTED).

15.3 Production Acceptance Checklist

- All 18 core UAT checks above marked PASS by the client, with evidence recorded.
- At least one full real day run side-by-side with the old billing method, reconciled to the rupee.
- Barcode scanner physically installed and the 5-minute scan check passed on the billing computer.
- One printed invoice and one printed label inspected for company name, GSTIN, values and code legibility.
- Backup and recovery reviewed with whoever hosts the database (see 15.4) and a restore actually practised once.
- Admin password stored safely; service key rotation completed; public sign-up confirmed disabled.
- This manual handed to every counter user with a 30-minute walkthrough of POS, returns and end-of-day.

15.4 Backup and Recovery Guidance

The database is the business record, so protect it in three layers. First, enable automated daily backups on the hosted database (Supabase paid tier includes point-in-time recovery; verify the retention window covers your comfort). Second, take a weekly manual export of key tables — sales, sale items, payments, stock movements, customers, suppliers, purchases, expenses — as CSV from the Reports exports and store a copy off the shop premises. Third, practise restoring: a backup that has never been restored is a hope, not a

backup. Recovery expectations to write down with your host: how far back point-in-time recovery reaches, and how long a full restore takes. The append-only stock ledger means partial recovery keeps the movement history trustworthy even if it must be re-summed.

15.5 Keeping This Manual Alive

This is a living document. Whenever a feature is added or changed, update its section in the same release: what it does, the user steps, the permission that gates it, any new setting, the error handling, the tester procedure with positive and negative cases, security notes, responsive behaviour, and hardware implications. A feature that ships without its manual entry is not finished. Future recommendations currently parked (not promised): Excel/CSV product import, bulk label printing, direct thermal printing, cash-drawer integration, opening/closing cash register sessions, approval limits for discounts, low-stock notifications, and GST accounting export — the full classified list lives in the Phase 7 audit report.